PACEVAL | MATHEMATICAL RESEARCH DEMONSTRATOR

Wind-Turbine Digital Twin Coefficient Functions with paceval

A verified original-SDK C++ study of reusable aerodynamic coefficients, power, torque, sensitivities and reduced-order rotor dynamics.



Cover artwork: original conceptual illustration. The scientific figures in this report are generated from actual paceval execution output, not measurements of a commercial turbine.

Evidence from actual run	Scope
31 reusable expression graphs and 308,791 paceval batch outputs	Real library execution
3,721 coefficient-map points, 4,800 operating samples and 720 time steps for each of two versions	Synthetic and deterministic
1,760 independent high-precision checks; CTest passed	Mathematical verification, not turbine certification

13 November 2026 | Full original-SDK execution | English research whitepaper

1. Industrial use case and current research context

Wind-turbine digital twins help investigate how the incident wind field, rotor speed, blade pitch, drivetrain loading and electrical constraints jointly influence operating performance. A reusable mathematical evaluation layer may let a researcher switch coefficient-function versions without rewriting the outer state integrator, and may offer a clear audit trail for changes to physical assumptions.

Modern reference environments such as NREL/OpenFAST combine aerodynamic, structural, control, electrical and, where relevant, hydrodynamic models. IEC 61400-12-1:2022 defines procedures for measuring wind-turbine power performance and evaluating uncertainty. These capabilities are broader than the mathematical submodel investigated here.

What this research project actually demonstrates

Part of industrial workflow	Implemented and verified in this package
Aerodynamic coefficient lookup	Two smooth, versioned synthetic Cp functions and one Ct function
Rotor and generator mathematical kernels	Aerodynamic torque, thrust, generated power and rating-consistent generator torque
Reduced-order temporal state	Rotor angular-velocity derivative and Euler update executed in paceval
Numerical scenario management	C++ provides wind, pitch, time-step ordering and output aggregation
Full aeroelastic coupled machine twin	Not implemented: requires dedicated data and validated tools

The practical contribution is a reproducible mathematical kernel suitable for subsequent substitution with real, licensed or measured coefficient tables; no field-measured power curve or turbine performance certification is claimed.

2. Defined research model and provenance

The original WP_042 proposal suggested the relationship P = 0.5 rho A v^3 Cp(lambda,beta) and a fixed rotor simulation with tabulated coefficient maps. The present implementation expands that starting point into explicit, independently testable functions and adds an interpolation baseline. Every numerical wind profile and coefficient used here is deliberately synthetic.

Parameter	Declared synthetic reference value / units
Rotor radius R	63 m, illustrative reference-scale rotor
Air density rho	1.225 kg/m³, constant
Rotor inertia J	4.0 × 10^7 kg m²
Drivetrain drag coefficient D	1.2 × 10^5 N m s
Generator reference torque and speed	2.3 MN m at 1.12 rad/s
Electrical efficiency and rated power	0.95, 5 MW
Euler time increment	1 s; dynamics integrated by C++ using a paceval next-state graph
Sweep domain	3 ≤ lambda ≤ 14 and 0° ≤ pitch ≤ 20°
Actual simulation wind speed	5.52 to 12.88 m/s over 720 seconds

These values are not a calibrated NREL 5-MW model; nominal size and rating alone do not reproduce a validated turbine. Some experimental operating points deliberately cover combinations outside normal turbine supervisory-controller schedules, because the purpose is mathematical function testing.

3. Aerodynamic coefficient mathematics

Tip-speed ratio is the nondimensional ratio of blade-tip speed to incoming wind speed. The rotor area is circular. The synthetic power coefficient uses a bounded Gaussian dependence on tip-speed ratio and an exponential decrease with blade pitch. Its second version adds a small, explicit, independently registered shape modification.

Explicit power coefficient

$$C_{p,1}(\lambda,\beta) = 0.475\exp[-((\lambda - 7.6)/3.1)^2]\exp(-\beta/22)$$

Version 2: Cp2 = Cp1 × [1 + 0.015 sin(0.5 lambda) exp(-beta/25)]. The thrust coefficient is Ct = 0.03 + 0.78 exp[-((lambda-7)/6)^2] exp(-beta/30). Both formulas are provided verbatim in the formula registry and as distinct paceval computation handles.

Bounds, sensitivities and what they mean

The two Cp models remain below the ideal Betz upper bound 16/27 throughout the tested domain. This is only a physical screening criterion. The model does not solve blade-element aerodynamics or guarantee realistic lift, drag, dynamic stall, tower shadow or yaw sensitivity.

The explicit analytical derivatives dCp/dlambda and dCp/dbeta are computed with the original paceval engine. The chain rule yields aerodynamic-power sensitivities at fixed omega, including dP/dv = (0.5 rho A v²) [3 Cp - lambda dCp/dlambda]. Avoiding a numerical derivative inside the runtime makes the mathematical assumptions auditable.

4. Power, thrust and generator-energy balance

Physical quantity definitions

$$P_a = \frac{1}{2}\rho\pi R^2 v^3 C_p, \quad Q_a = \frac{P_a}{\omega}, \quad F_t = \frac{1}{2}\rho\pi R^2 v^2 C_t$$

Aerodynamic rotor torque is obtained by dividing aerodynamic power by positive rotor angular speed. Axial force uses a separately declared thrust coefficient. The tested domain excludes zero wind speed and zero rotor speed, where these simplified relations would require explicit treatment of startup, zero torque and aerodynamic nonlinearities.

Rating-consistent electrical extraction

$$S(e) = \frac{1}{2}[e + P_r - \sqrt{(e - P_r)^2 + \varepsilon^2}], \quad Q_g = \frac{S(e)}{\eta\omega}$$

The generator reference law first produces an unconstrained electrical power proxy. A smooth rated-power limit S then determines both the delivered power and the corresponding shaft reaction torque. This is important: separately clipping output power while retaining the unconstrained shaft torque would violate the declared energy balance.

The smooth rating proxy uses epsilon = 1,500 W and is not an inverter or brake controller. The generator electrical loss is S(e) × (1/eta - 1) and is computed in a separate paceval graph. Rated-power margins are also independently available as reusable expression outputs.

State equation and explicit Euler update

$$J\dot{\omega} = Q_a - Q_g - D\omega, \qquad \omega_{n+1} = \omega_n + \Delta t\dot{\omega}_n$$

The full analytical Euler next-state expression is executed with paceval for both coefficient versions; C++ only advances the state, validates the domain, and sums electrical energy over chronological time steps.

5. Original paceval SDK: architecture and execution

The application compiles the original paceval_main.cpp and paceval_main.h files, links an independently supplied original shared library, and creates 31 persistent computation handles from literal expressions. On each evaluated input block it invokes paceval_dGetMultipleComputationsResultsExt and checks per-result errors and finite outputs.

Graph family	Real paceval graph outputs
Coefficients and geometry	lambda, Cp v1, Cp v2, Ct, rotor area, wind incident power
Aerodynamic force and power	rotor power v1/v2, rotor torque v1/v2, thrust
Analytical sensitivities	dCp/dlambda, dCp/dbeta, dP/dv, dP/domega, dP/dbeta
Generator and operating envelope	reaction torque, mechanical/electrical power, smooth rating, losses, Betz/rated/wind margins
Dynamics and versioning	angular acceleration v1/v2, Euler next-state v1/v2, delta Cp and delta power

The mandatory machine-readable results/math_execution_inventory.csv lists each executed expression, owner and API path. The C++ host retains responsibility for stateful integration, wind and pitch profile generation, numeric aggregation, input validation and data export.

Because the dynamical system is stateful, batch evaluation does not make the entire simulation simultaneous: 31 graphs are still batch-evaluated per step, while the time update is deliberately sequential. The coefficient maps and operating sweeps use contiguous batched parameter arrays.

6. Actually executed and independently validated results

Executed metric	Verified run result
Original paceval persistent mathematical graphs	31
Successful multi-graph batch calls	788
Successfully evaluated batch graph outputs	308,791
Coefficient-map mesh	61 × 61 = 3,721 points
Additional parameter sweep	4,800 points
Sequential rotor traces	2 versions × 720 samples
Independent C++ oracle values	22,841 passed
Independent 80-digit mpmath reference values	1,760 passed
Finite-difference / dynamical reference checks	80 / 7,200 passed
Version 1 generated energy	508.646 kWh (synthetic)
Version 2 generated energy	504.387 kWh (synthetic)
Build and integration test	CMake succeeded; CTest 1/1 passed

The numeric run proves that the specified reusable formula registry executes consistently. It does not establish the validity of any real turbine production forecast, power curve, safety envelope, fatigue loading or aerodynamic model.

7. Computed aerodynamic Cp and rotor-power surfaces

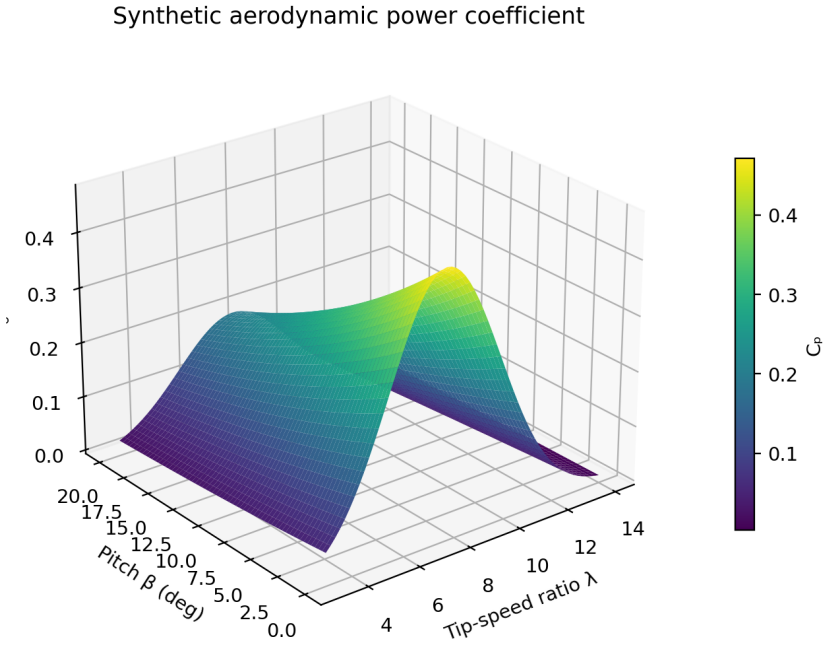


Figure 1. Power coefficient Cp over tip-speed ratio and pitch, calculated with paceval.

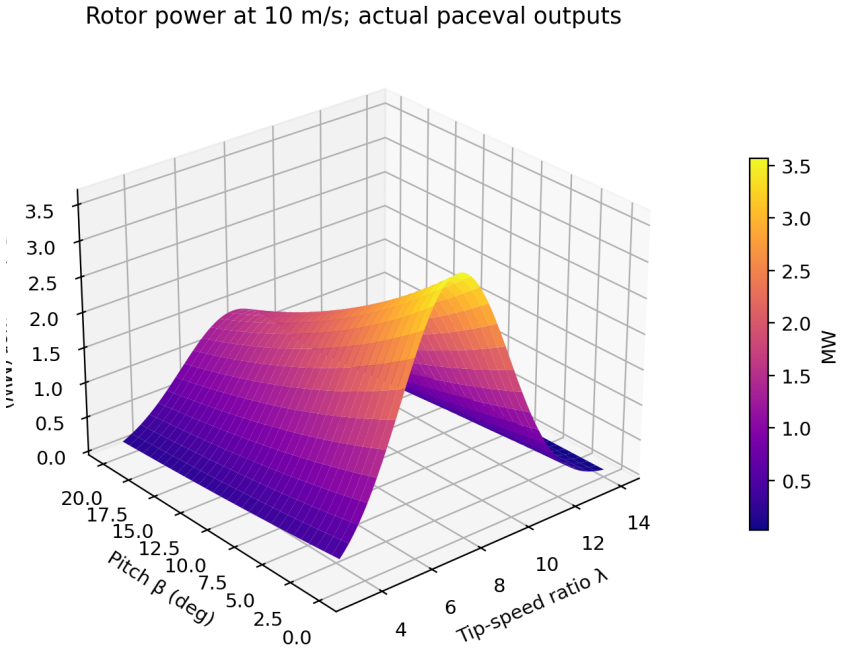


Figure 2. Rotor power at 10 m/s, directly from the same paceval aerodynamic graph family.

8. Coefficient and thrust operating envelopes

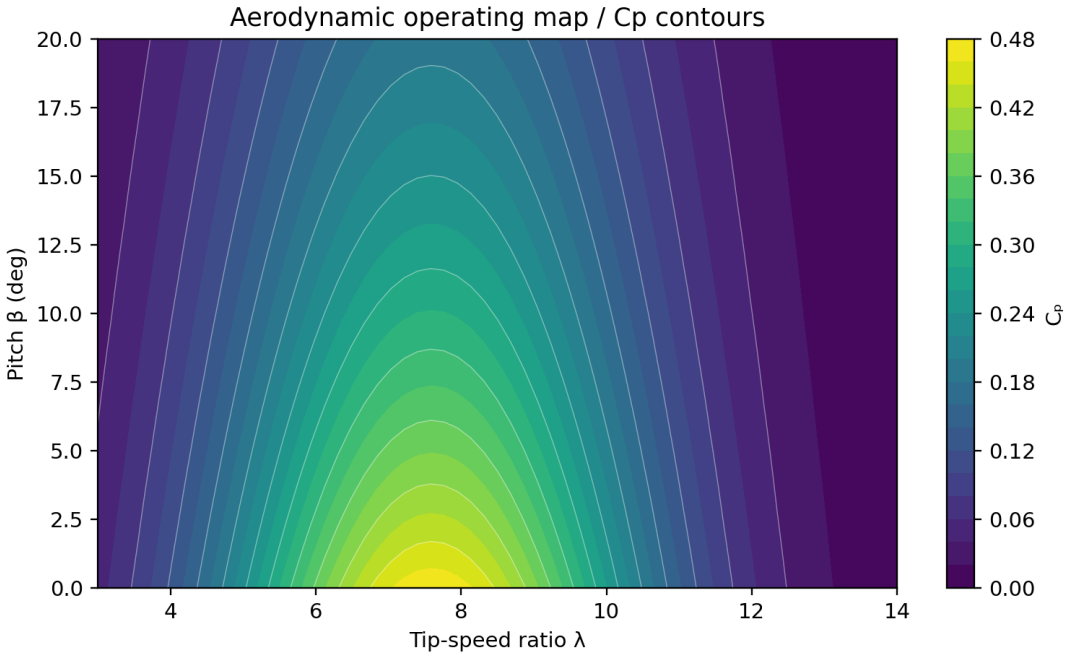


Figure 3. Cp isolines and model operating region; synthetic coefficient, not a measured curve.

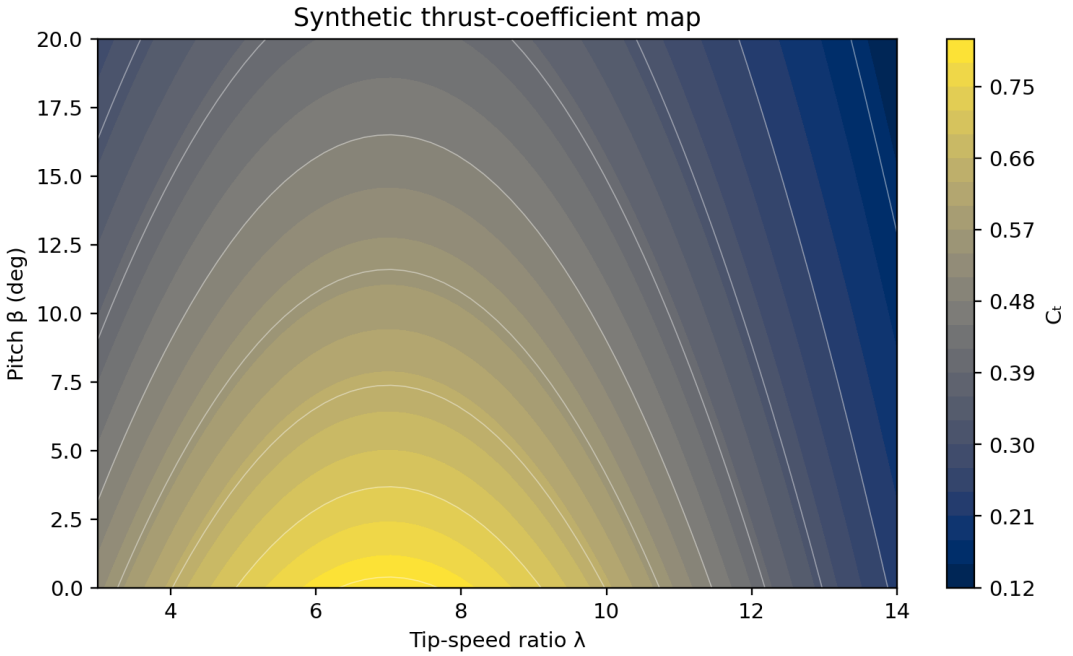


Figure 4. Axial-thrust coefficient Ct as a function of the two controlled inputs.

9. Analytical derivatives and formula versioning

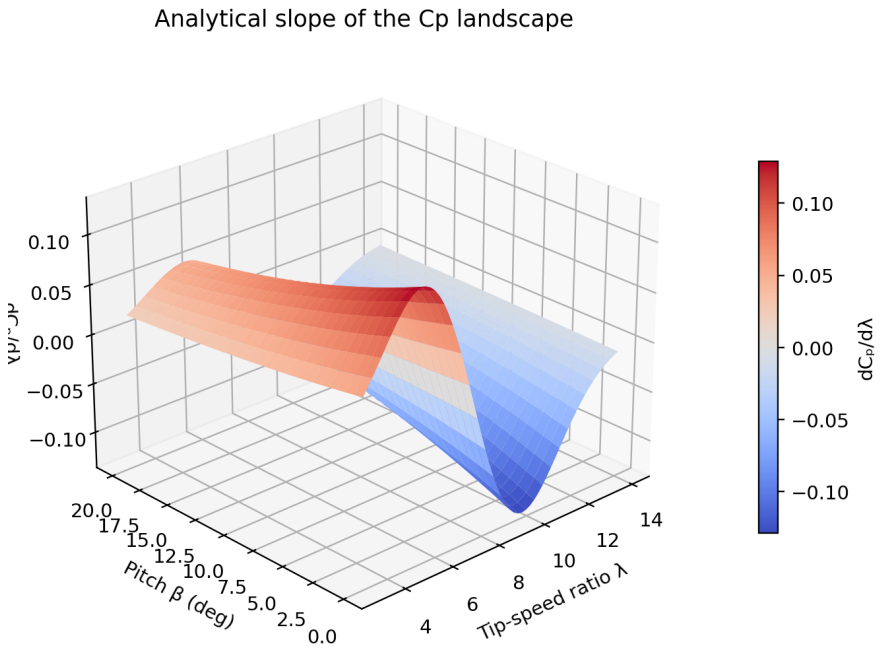


Figure 5. Analytical dCp/dlambda surface, evaluated as a reusable paceval graph.

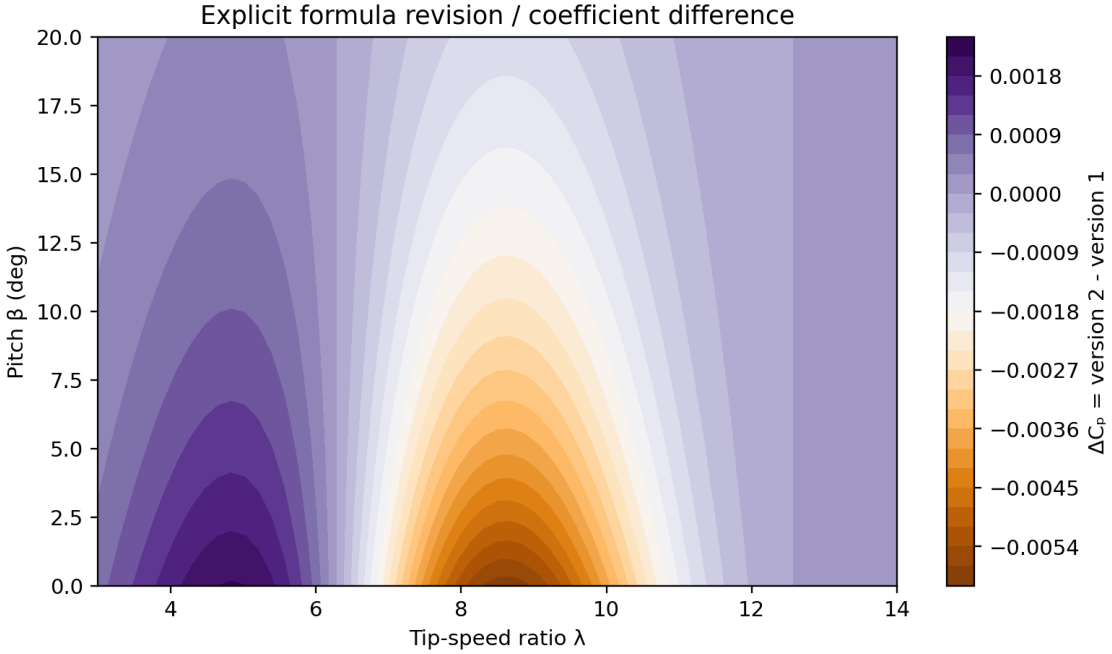


Figure 9. Explicitly versioned coefficient difference Cp(v2) minus Cp(v1).

10. Reduced-order dynamic rotor experiment

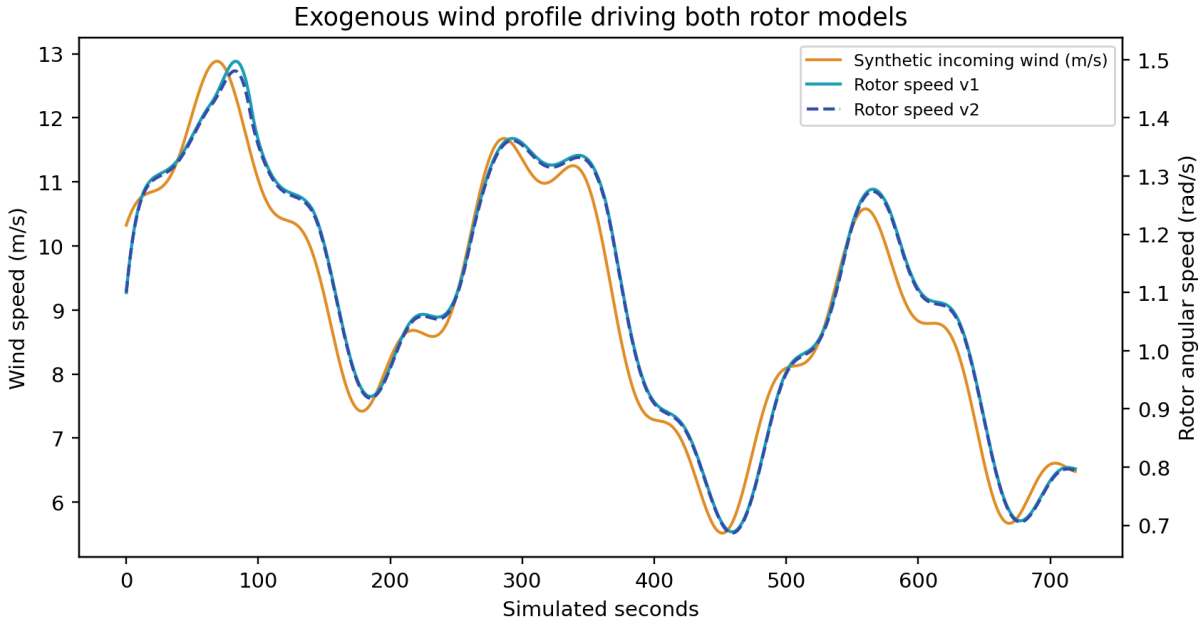


Figure 6. Reproducible synthetic wind speed and two corresponding angular-velocity traces.

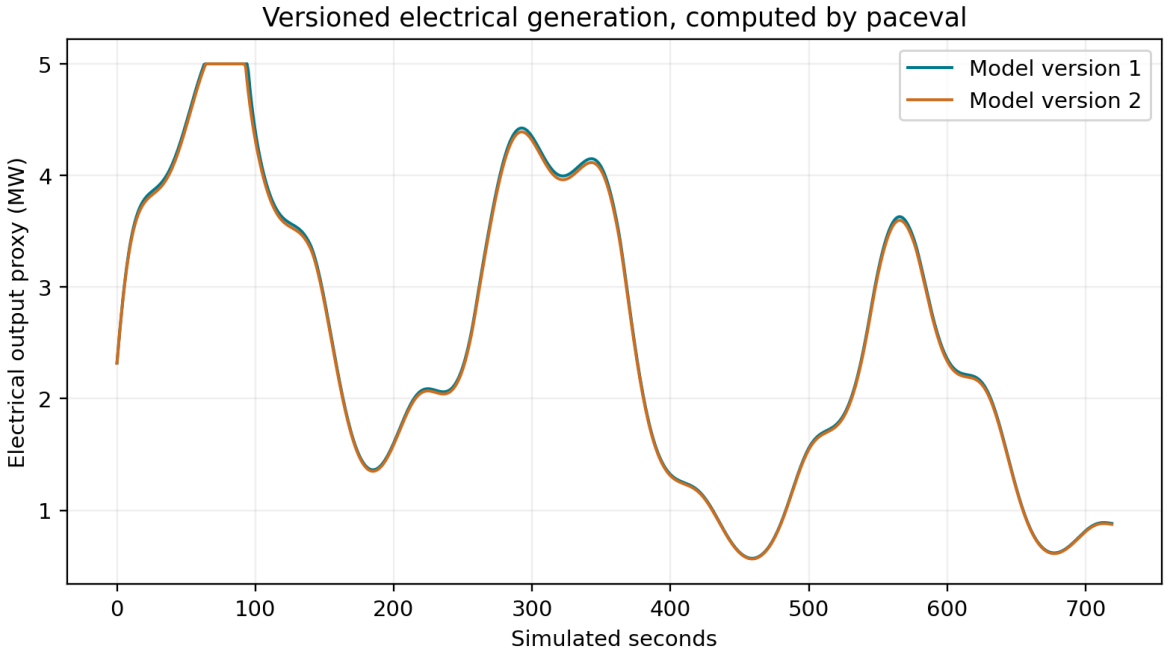


Figure 7. Rate-limited electrical power proxies evaluated by paceval during sequential rotor integration.

11. Energy accumulation and operating-point survey

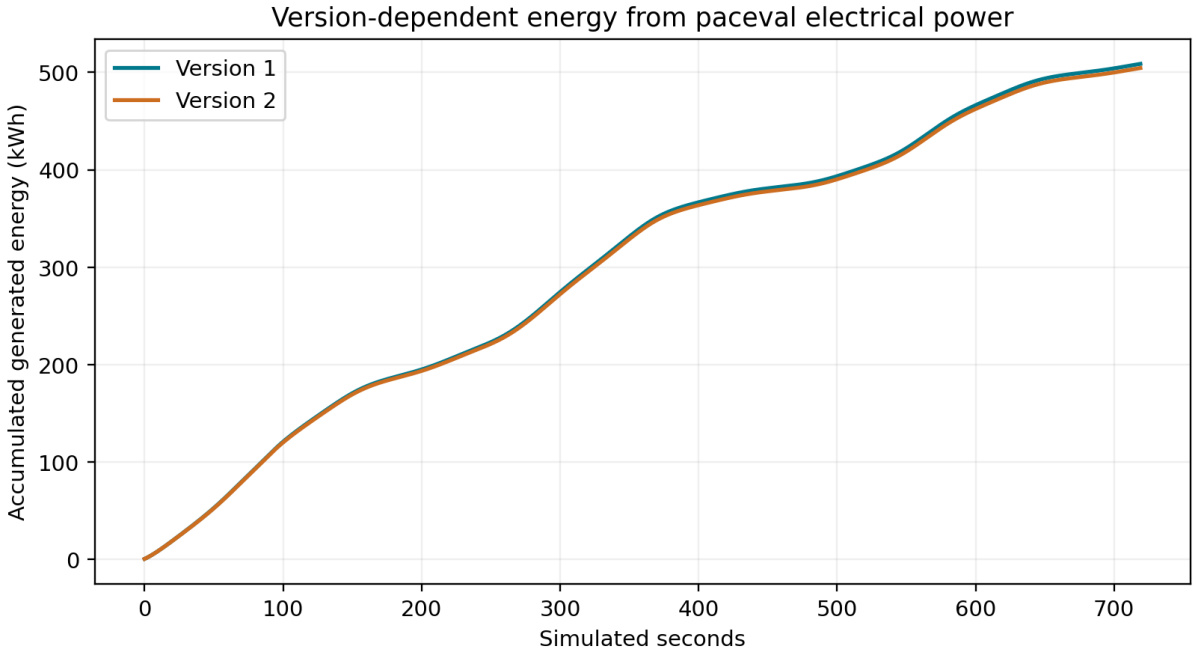


Figure 8. Host-accumulated energy from actual paceval electrical-power time series (two versions).

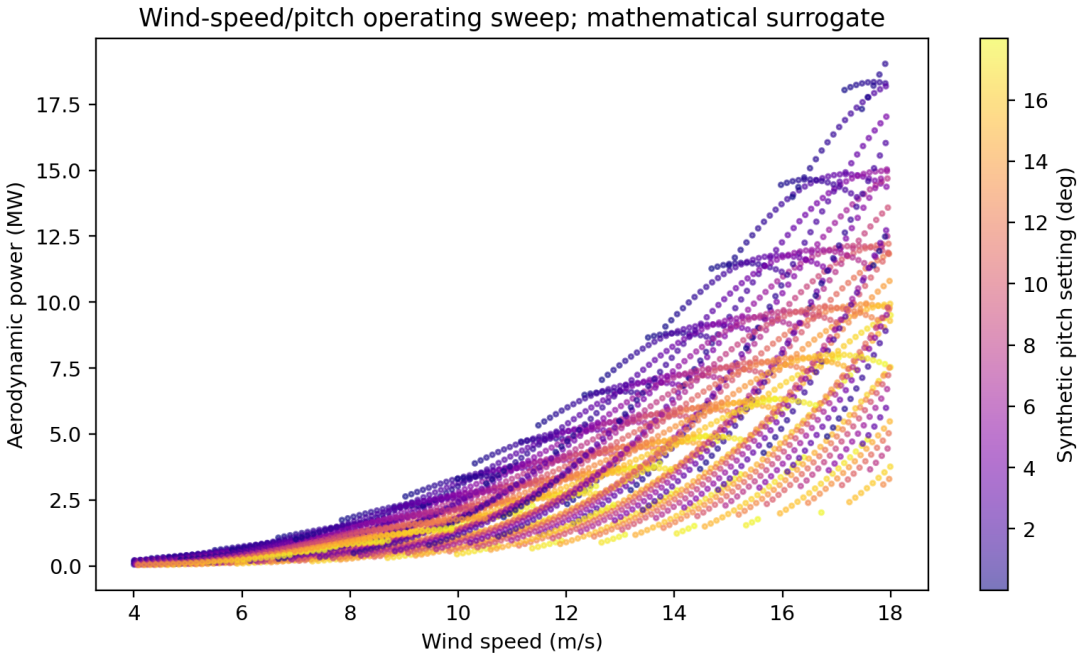


Figure 10. Four-thousand-eight-hundred synthetic wind, pitch and rotor-speed combinations.

12. Independent coefficient table and error audit

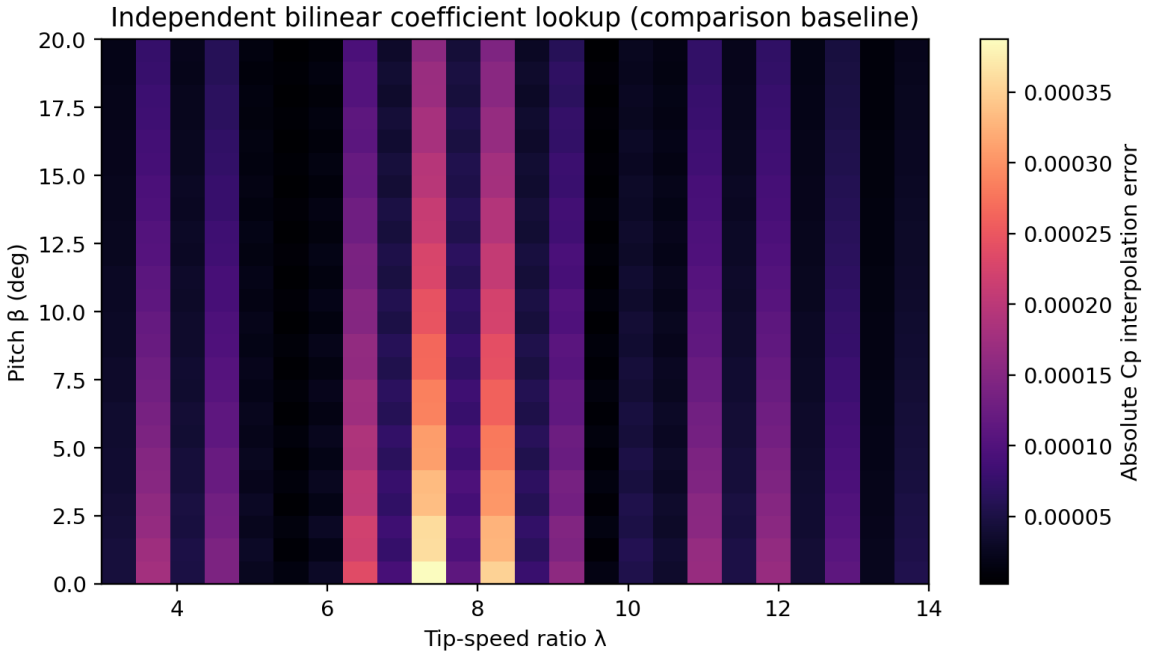


Figure 11. Independent bilinear coefficient-map interpolation versus the explicitly declared synthetic Cp formula.

The comparison evaluates 576 off-grid points against the 61 × 61 pre-tabulated paceval Cp surface. The maximum absolute interpolation difference is 0.000387, and the mean difference is 0.000072. This is a numerical approximation check of a smooth surrogate, not measured aerodynamics.

An independent 80-decimal-digit oracle passed 1,760 values; numerical checks of 80 analytical finite differences and 7,200 dynamic relations also passed. The maximum reported scaled reference discrepancy in the 80-digit sample was 2.59e-13.

13. Scientific interpretation and limitation boundaries

What the executable demonstrator establishes

The original paceval engine evaluates reusable nonlinear mathematical coefficient functions and composite quantities across a large synthetic parameter domain, and it can be used inside a stateful C++ integrator. The two versioned coefficient functions remain available as independently stored expression graphs. Mathematical responsibilities are traceable and the results are reproducible.

What remains outside the tested scope

The numerical study omits real SCADA data, measured or BEM-derived coefficient tables, turbulence and wake fields, blade deformation, dynamic stall, drivetrain nonlinearities, certified protection functions, structural loads, fatigue damage and uncertainty propagation. Simplified Cp and Ct functions and a smooth generator cap cannot substitute for an OpenFAST aero-hydro-servo-elastic simulation.

The reference IEC 61400-12-1:2022 is a power-performance measurement standard, not a certification achieved by this work. These simulation outputs have neither statistical uncertainty bands from physical measurements nor extrapolation guarantees. A smaller difference between two formula versions does not demonstrate a useful real-world improvement.

Next realistic research milestone

Replace the explicitly synthetic coefficient curves with correctly licensed NREL/OpenFAST BEM or turbine-specific measured coefficient data, freeze a valid operating envelope, and test independent wind and pitch profiles with uncertainty-aware verification against the same rotor integrator. Any claim about operational improvement requires appropriate real measurements and reproducible comparisons.

14. Reproducibility, references and conclusion

Build the C++ executable using a compatible original external paceval Linux runtime, keeping the included original header and interface source. Execute the 720-step study and 61-by-61 coefficient map, regenerate the independent Python numerical audit, scientific figures and this PDF, then run the README link and archive checks.

Build: cmake -S src -B build -DCMAKE_BUILD_TYPE=Release -DPACEVAL_RUNTIME=/absolute/path/libpaceval_linux_sharedLIB.so

Execute: cmake --build build -j; ctest --test-dir build --output-on-failure; ./build/wind_turbine_twin --out . --steps 720 --surface 61 --sweep 4800

Reproduction files include examples/formula_registry.csv, the full coefficient and operating surfaces, versioned dynamic traces, results/math_execution_inventory.csv, high_precision_validation.json and SDK execution log. The complete original library binary is not redistributed.

Selected scientific sources

[1] NREL / OpenFAST v5.0.0 (November 6, 2026), multi-physics turbine dynamics; https://openfast.readthedocs.io/en/dev/

[2] IEC 61400-12-1:2022, power performance measurements; https://webstore.iec.ch/en/publication/68499

[3] NLR Wind Systems WISDEM reference turbine 5MW YAML; https://github.com/NLRWindSystems/WISDEM/blob/master/examples/02_reference_turbines/nrel5mw.yaml

[4] Original paceval C++ mathematical engine and public SDK, https://github.com/paceval/paceval

[5] FMI standard documentation, https://fmi-standard.org/docs/main/

[6] WP_042 original independent five-page user proposal (reproduced in the project).

Conclusion

Successful mathematical execution proof: 31 real original-SDK graphs, more than 300,000 verified batch outputs, an energy-consistent reduced-order rotor update, and independent high-precision checks demonstrate reusable mathematical turbine coefficient functions. Physical turbine validation and a full operational digital twin remain future work.